PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab UniversityAct VII of1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (HOSPITAL ADMINISTRATION AND MANAGEMENT)

As per NEP 2020

w.e.f session 2026-27

Course Outline for B.Voc (HOSPITAL ADMINISTRATION AND MANAGEMENT) as per NEP 2020

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 101	Core Discipline	Introduction to Healthcare Industry	4	-	4	80	20	-	-	100
HAM 102	Core Discipline	Principles of Management	4	-	4	80	20	-	-	100
HAM 103	Core Discipline	Principles Of Marketing	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	Common Basket			3					75
IDC I	Interdisciplinary Course	Common Basket			3					75
VAC I	Value-Added Course	Common Basket			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 201	Core Discipline	Hospital Management Information System	4	-	4	80	20	-	-	100
HAM 202	Core Discipline	Managing Quality in Healthcare	4	-	4	80	20	-	-	100
HAM 203	Core Discipline	Organisational Behaviour	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	Common Basket			3					75
IDC II	Interdisciplinary Course	Common Basket			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Patient Care Coordinator

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training -report by the student to be submitted within week of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Hospital Administration and Management= 48 + 4 #.

#Exit 1: Certificate Course- Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 301	Core Discipline	Human Resource Management	4	-	4	80	20	-	-	100
HAM 302	Core Discipline	Hospital Planning & Designing	4	-	4	80	20	-	-	100
HAM 303	Core Discipline	Customer- Relationship Management	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 401	Core Discipline	Community Health & Management Of National Health Programmes	4	-	4	80	20	-	-	100
HAM 402	Core Discipline	Introduction To Epidemiology & Bio Statistics	4	-	4	80	20	-	-	100
HAM 403	Core Discipline	Hazards And Risk Management	2	2	4	40	10	40	10	100
HAM 404	Core Discipline	Financial Management	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Patient Relations Executive

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-voce examination to be held within 3 weeks of the start of 3rd semester.

**Minimum Credits required to get Diploma Course in Hospital Administration and Management= 96 + 4 #.

#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 501	Core Discipline	Medicals Ethics And Legal Issues In Hospital	4	-	4	80	20	-	-	100
HAM 502	Core Discipline	Hospital Materials Management	4	-	4	80	20	-	-	100
HAM 503	Core Discipline	Market Research And Consumer Behaviour In Healthcare	2	2	4	40	10	40	10	100
HAM 504^	Core Discipline	Strategic Management In Healthcare	4	-	4*	80	20	-	-	100
HAM 505^	Core Discipline	Patient Care & Healthcare Technology	4	-	4*	80	20	-	-	100
HAM 506^	Core Discipline	Brand Management	4	-	4*	80	20	-	-	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 601	Core Discipline	Hospital Administration	4	-	4	80	20	-	-	100
HAM 602	Core Discipline	Nursing Administration and Management	4	-	4	80	20	-	-	100
HAM 603	Core Discipline	Biomedical Waste Management	2	2	4	40	10	40	10	100
HAM 604	Core Discipline	Hospital Operations Management	2	2	4	40	10	40	10	100
HAM 605^	Core Discipline	Medical Terminology And Procedures	4	-	4*	80	20	-	-	100*
HAM 606^	Core Discipline	Research Methodology	4	-	4*	80	20	-	-	100*
HAM 607^	Core Discipline	Management & Organisation Of Medical Records	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

Job Role-Assistant Hospital Administrator

Semester-VII

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 701	Core Discipline	Healthcare Economics	4	-	4	80	20	-	-	100
HAM 702	Core Discipline	Health Insurance	4	-	4	80	20	-	-	100
HAM 703	Core Discipline	Healthcare Delivery System	2	2	4	40	10	40	10	100
HAM 704^	Core Discipline	International Classification Of Diseases	4	-	4*	80	20	-	-	100*
HAM 705^	Core Discipline	Total Quality Management Tools And Techniques	4	-	4*	80	20	-	-	100*
HAM 706^	Core Discipline	Healthcare In India	4	-	4*	80	20	-	-	100*
SIT 701	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

^Any two in 7th Semester

Semester-VIII

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HAM 801	Core Discipline	Service Marketing	4	-	4	80	20	-	-	100
HAM 802	Core Discipline	Business Environment	4	-	4	80	20	-	-	100
HAM 803	Core Discipline	Operation Research & Management	2	2	4	40	10	40	10	100
HAM 804	Core Discipline	Hospital Facilities Management	2	2	4	40	10	40	10	100
HAM 805^	Core Discipline	Bio Medical Instrumentation	4	-	4*	80	20	-	-	100*
HAM 806^	Core Discipline	Newer Developments in Hospital Administration	4	-	4*	80	20	-	-	100*
HAM 807^	Core Discipline	Drug Management	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 8th Semester

Job Role –Hospital Administrator

*Summer Industrial Training of 2-4weeks in a relevant Industry after 6th Semester Examinations during summer break. Training- report by the student to be submitted within one week of start of 7th Semester. Viva-Voce examination to be held within 3 weeks of the start of 7th semester.

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (HOSPITAL ADMINISTRATION AND MANAGEMENT) (SEMESTER I - VIII)

PROGRAMME OUTCOMES

PO1. To create professional and effective managers of hospitals with leadership and problem-solving skills.

PO2. To attain a general level of competence in hospital management and to enable them to act with creative, innovative and entrepreneurial potentials with management tools.

PO3. To provide advancement of conceptual and practical knowledge in the field of hospital management to contribute to nation building while upholding ethical practices.

PO4. To develop competent management professionals with strong ethical values capable of assuming a significant role in Corporate and Society**.**

PO5. To enable the students possess professional skills for employment and lifelong learning in management.

Semester I

1	Course Title	Introduction to Healthcare Industry
2	**Course Code**	**HAM 101**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of this course is to help students learn and understand the Healthcare and Hospital management so that they can get an overview and get oriented toward healthcare and hospital management.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Will be able to gain insights on meaning and significance of healthcare sector. CO2. Will be able to get knowledge on the role of hospitals. CO3. Will be able to get in-depth knowledge on hospital administration. CO4. Get introduced to various supportive services. CO5. Gain knowledge on hospitals management roles and responsibilities.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

UNIT – I

Meaning and importance of healthcare and hospital management, Disease, care and administration, Role and place of hospital in health spectrum, health indicators and their importance.

UNIT – II

Hospital Sector: definition, meaning, and significance of Hospital sector, functions and classification of hospitals, role of hospitals in society.

UNIT – III

Health administration in India: An overview of medical care, including the role of state and local self-government, NGOs, Private and corporate sector, community at large. Organizational structure of hospitals, services including clinical, administrative and supportive services managed by the hospitals.

UNIT – IV

Hospital management: levels and roles, governing bodies, executive board, advisory board, CEO, medical administration, nursing administration and hospital administration, middle level managers in the hospitals and their responsibilities.

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13 Suggested Resources for Reading

1. 1. Joshi, D.C., *Hospital Administration*, Jaypee Brothers**.**
2. Dunn & Haimann's Healthcare Management, Rose. T. Dunn Health Administration Pr; 9th Edtion-2010.
3. Hospital Management and Administration Principles and Practice Including Law, BV Subramanyam, CBS Publishers& Distribution Pvt Ltd.
4. Management Principles for Health Professionals, Joan Gratto Liebler, Charles R.Mc Connell, Jones and Bartett Publishers, Inc, 7th Edition-2016.
5. Hospital Administration and Management –A Comprehensive Guide, Joydeep Das Gupta, Jaypee Brothers Medical Publishers; Second Edition-2015.
6. Hospital Management; Text & cases, Pearson Education India, First Edition-2013.

Note: Latest edition of text books may be used.

1	Course Title	Principles Of Management
2	**Course Code**	**HAM 102**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of this course is to provide in-depth knowledge of various management concepts and its utility in emerging scenario.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Will be able to gain insights on different management concepts and their significance on healthcare sector. CO2. Will be able to get knowledge on planning and its process. CO3. Will be able to get in-depth knowledge on organizing and its structure. CO4. Get introduced to staffing functions and its various types, role of direction and importance and types of performance appraisal . CO5. Gain knowledge on various theories of motivation its role in employee performance with the significance of controlling function**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

UNIT – I

MANAGEMENT: Concept –Nature- Levels- Skills- Functions - Management as an Art, Science and Profession – F.W.Taylor's Scientific Management Theory - Fayol's Theory of Management- Human Relations Approach; Healthcare Management: Significance- Role and Responsibility of Healthcare Manager in Hospital – Corporate Social Responsibility.

UNIT – II

sPLANNING: Concept-Nature-Purpose-Process of Planning-Types of Plans – Premising; Decision Making: Concept- Decision Making Processes; Management by Objectives: Concept Process. ORGANIZING: Nature-Purpose-Process-Formal and Informal Organizations Departmentation- Span of Control-Delegation-Decentralization-Line and Staff - Committees.

UNIT – III

STAFFING: Nature and Importance of Staffing – Recruitment - Selecting - Training and Development - Performance Appraisal; DIRECTING: Meaning-Nature and importance Principles of effective Directing-Assumptions of Human Behaviour by Douglas McGregor, Edgar Shien - Elton Mayo.

UNIT – IV

MOTIVATION: Concept – Significance - Theories of Motivation - Leadership - Leadership Styles-Leadership Theories; Communication: Concept -Importance-Process-Barriers Principles of Effective Communication; CONTROLLING: Concept- Pre-Requisites and characteristics of effective control systems- Basic control process - Controlling Techniques**.**

13 Suggested Resources for Reading

1. Principles and Practices of Management, LM Prasad, Sultan Chand& Sons, New Delhi2019.
2. Management An Introduction, David Boddy, Pearson Education, Seventh Edition- 2018.
3. Essentials of Hospital Management & Administration, D.H.L. Ramachandra, Edu creation Publishing2018.
4. Principles of Management, 6th Edition, PC Tripathi and PN Reddy, McGraw Hill Education-2017**.**

Note: Latest edition of text books may be used.

1	Course Title	Principles of Marketing
2	**Course Code**	**HAM 103**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH +40PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main aim of the paper is to understand the nature, tasks and environment under which marketing operates, to study the theory, principles and practical aspects of various marketing functions, and to learn to take marketing decisions.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. State the role and functions of marketing within a range of organizations. CO2. Describe key marketing concepts, theories and techniques for analyzing a variety of marketing situations. CO3. Identify and demonstrate the dynamic nature of the environment in which marketing Decisions are taken and appreciate the implications for marketing strategy determination and implementation. CO4. Apply the introduced conceptual frameworks, theory and techniques to various marketing contexts. CO5. Analyze the relevance of marketing concepts and theories in evaluating the impacts of environmental changes on marketing planning, strategies and practices. CO6. Demonstrate the ability to carry out a research project that explores marketing planning and strategies for a specific marketing situation**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

UNIT – I

Introduction to Marketing: Definition, Scope and Importance of Marketing, Key Customer Markets, Concepts/Philosophies of Marketing, Holistic Marketing Concept, Marketing Tasks, Marketing Mix Marketing Environment: Marketing Environment, New Marketing Realities, New Consumer Capabilities, Demographic Environment, Social-Cultural Environment, Natural Environment, Technological Environment and Political-Legal Environment, SWOT analysis, Role of Marketing in Health Care Organizations.

UNIT – II

Analyzing Markets: Marketing Research Process, Sources of data collection, factors influencing Consumer Behaviour, Buying Decision Process, Post-Purchase Behavior, Organizational Buying, and Stages in the Buying Process. Market Segmentation: Levels of Market Segmentation, segmenting consumer markets, Niche Marketing ,segmenting business markets, Michael Porter's Five Forces Model, Analyzing competitors, strategies for market leaders, Targeting and Positioning.

UNIT – III

Product Decisions: Product and Services characteristics, classifications, differentiation, packaging and labeling, branding, Product Life Cycle. Pricing Strategies: Understanding Pricing, Setting the Price, Initiating and Responding to Price Changes, and Reactions to Competitor's Price Changes**.**

UNIT-IV

Marketing Channels: Marketing Channels, Role of Marketing Channels, Identifying Major Channel Alternatives, Types of Intermediaries, Channel-Management Decisions, Retailing, 7 Wholesaling. Managing Conflicts and Controls in Channels, channel Integration and Systems and Logistics. Marketing Communication: The Role of Marketing Communications, Communications Mix-Advertising, Sales Promotion, Public Relations and Publicity, Events and Experiences, Direct and Interactive Marketing, Personal Selling.

13 Suggested Resources For Reading

1. Kotler, P. and Clarke, R. N., *Marketing for Healthcare Organizations,* Prentice Hall Publication.
2. Silver, R., *Health Service Public Relations,* Radcliffe Medical Press Ltd.
3. Kunder, G. D., *How to Market Your Hospital Without Selling Your Philosophy,* Prism Books Pvt. Ltd.
4. Kotler, P., Shalowitz, J. I., and Stevens, R. J., *Strategic Marketing for Health Care eOrganizations: Building a Customer-Driven Health System*. John Wiley and Sons.
5. Thomas, R.K. and Calhoun, M., *Marketing Matters: A Guide for Healthcare Executives*. PHI Learning Pvt. Ltd.
6. Berkowitz, E. N., *Essentials of Health Care Marketing,* Jones and Bartlett Learning**.**

Note: Latest edition of text books may be used.

1	Course Title	Project Work-I and Viva-Voce
2	**Course Code**	**PW1**
3	**Typology of Course**	Project work
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	2 Credits
6	**Max. Marks**	50 Marks
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of the /Project Work Course is to help the student develop his/her ability to apply multi-disciplinary concepts, tools and techniques to solve organizational problems and/or to evolve new/innovative theoretical frame work.
9	**Course Content The Project may take any one of the following forms:** **(i)** Comprehensive case study (covering single organization/multifunctional area problem, formulation analysis and recommendations) **(ii)** Inter-organizational study aimed at inter-organizational comparison/ validation of theory/survey of management services. **(iii)** Evolution of any new conceptual / theoretical framework. **(iv)** Field study (emperical study)	

Semester 2

1	Course Title	Hospital Management Information System
2	**Course Code**	**HAM 201**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	**Objectives**: To make the learner understand MIS as a managerial decision-making tool and to know the sources and compiling of MIS.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Know the principles of total quality management and peculiarities of their implementation. CO2. Know prerequisites of evolution of total quality management and significance of quality gurus' works to the management of modern organizations. CO3. Be able to use quality management methods analyzing and solving problems of organization. CO4. Choose a framework to evaluate the performance excellence of an organization, and determine the set of performance indicators that will align people with the objectives of the organization**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory ,Interactive sessions ,PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

UNIT – I

The Information Explosion: Information is important – Impact on society – Impact on teaching and learning – Impact on Government – Impact on Healthcare – The future of healthcare technology – The future healthcare record – Preparing for the future – Summary. The world of Informatics.

UNIT – II

The Electronic health record: Functions of the health record –Changing functions of the patients record – Advantages of the paper record – Disadvantages of the paper record – Optically scanned records – The electronic health record – Automating the paper record – Advantages of the EHR – Disadvantages of the EHR – Bedside or point-or-care systems – Human factors and the EHR – Roadblocks and challenges to EHR implementation –The future.

UNIT – III

Securing the Information: Privacy and confidentiality and Law – Who owns the data? – Security – Computer crime – Role of healthcare professionals – Summary. Information Systems cycle: The information systems cycle – Analysis – Design phase – Development – Implementation – Why some projects fails? Electronic Communications: A bit of history – Hardware and software for connecting – Methods of accessing information – World Wide Web (WEB) – Communication Technologies.

UNIT – IV

Telehealth– Historical perspective on telehealth – Types of Technology – Clinical initiatives – Administrative initiatives – Advantages and Barriers of telehealth – Future trends – Summary- The future of Informatics: Globalization of Information Technology – Electronic communication – Knowledge management – Genomics – Advances in public health – Speech recognition – Wireless computing – Security – Telehealth – Informatics Education – Barriers to Information Technology implementation

13 Suggested Resources for Reading

1. O'Brien, J., *Management Information System*, Tata Mc Graw Hill
2. Norton, P., *Introduction to Computer*, Tata Mc Gra w Hill
3. Green, M. A. and Bowie, M. J. O., *Essentials of Health Information Management: Principles and Practices*, Thomson Delmar Learning.
4. Kolker, A., *Healthcare Management Engineering*, Springer.
5. Jan, F., *Step by Step Quality Care*, Jaypee Brothers**.**

Note: Latest edition of text books may be used.

	Course Title	Managing Quality in Healthcare
2	**Course Code**	**HAM 202**
3	**Typology of Course**	Core Discipline
14	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees'
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	This course intends to familiarize and develop an understanding of the students with the basic principles and practices of quality in healthcare to administer hospitals and deliver healthcare services efficiently**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Know the principles of total quality management and peculiarities of their implementation. CO2. Know prerequisites of evolution of total quality management and significance of quality gurus' works to the management of modern organizations. CO3. Be able to use quality management methods analyzing and solving problems of organization. CO4. Choose a framework to evaluate the performance excellence of an organization, and determine the set of performance indicators that will align people with the objectives of the organization.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory ,Interactive sessions ,PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Quality in Healthcare: Dimensions of quality, parameters to improve quality in healthcare, TQM, Lean Thinking, Six Sigma, Kaizen, Quality Circles.

UNIT – II

Certification and Accreditation in Hospitals: ISO, NABH, JCI, JCAHO Programs for Hospital-wide Improvement of Services: Statutory Compliance in Hospitals, Patient Safety Management Program, Disaster Management Program in Hospital, Training Program in Hospital, Patient's Information and Education Program, Rights and Responsibilities of Patients.

UNIT – III

Quality Management at the Departmental Level: OT, Blood Bank, Inpatient Areas, Intensive Care Unit, CSSD, Medical Gases, Fire Safety, Outreach Services, and Dialysis Service.

UNIT – IV

Evaluation of Performance: Approaches to Evaluation of the QMS, Evaluation through Statistical Approach, Evaluation through Medical Audit, Evaluation through the Nursing unit, Equipment Audit, and Patient Satisfaction Survey.

13 Suggested Resources for Reading

1.Ellis, R. and Whittington, D., *Quality Assurance for Health Care: A Hand Book,* CRC Press
2. Joshi, S.K., *Quality Management in Hospitals*, Jaypee Brothers.
3. Evans, J. R., *Total Quality Management*, Language Learning.
4. Love, T., *Healthcare System Quality Improvements: Hospital Assessments*, Kindle Edition.

Note: Latest edition of text books may be used.

1	Course Title	Organizational Behavior
2	**Course Code**	**HAM 203**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40TH+40PR +20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of studying Organizational Behavior (OB) is to understand the underlying assumptions and causation of the individual's behavior and its impact on achieving the organizational goals.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. To identify the impact of individual behavior on groups and on the organization as a whole. CO2. To analyze and compare various theories of personality and learning theories. CO3. To identify the role and importance of communication and resolving conflicts. CO4. To explain group dynamics and demonstrate skills required for working in groups (team building). CO5. To identify the various leadership styles and the role of leaders in a decision making process. CO6. To analyze and compare different models used to explain individual behavior related to motivation and rewards.
10	**Pedagogy of the Course Work**	Class Room Lecture ,Laboratory, Interactive sessions, PPT, Practical and Group Discussions ,Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Introduction to OB: Contributing fields to OB, OB in the context of globalization. Learning and Behavior Management: Biographical characteristics, ability, job fit, learning theories, the process of behavior modification. Values, Attitudes and Job Satisfaction: The Nature and Dimensions of Attitudes, Components of Attitude, Sources and types, Cognitive dissonance theory, Values, Job satisfaction. Perception: Factors Influencing Perception, Perceptual Selectivity, Attribution Theory, Errors in Perception, Rational decision-making model, Linkage between Perception and Individual Decision Making

UNIT – II

Personality: The meaning of Personality, Personality Determinants, Personality Traits, The big five model, Emotional Intelligence. Motivation: Meaning of Motivation, Primary, General Motives and Secondary Motives, Motivation and Productivity, Content and process theories of Motivation. Group Dynamics: Types of Groups, Stages of Group Development: The Five Stage Model, The Punctuated Equilibrium Model, Teams in the Modern Workplace, Teams vs Groups - Types of Teams, Group decision making. Communication: Role of Communication, Objectives, Barriers to Effective Communication, Communication Processes, Types of Communication, Interactive Communication in Organizations.

UNIT – III

Leadership: Understanding Leadership, Leadership Theories - Trait Theories, Behavioral Theories, Contingency Theories, Leadership Styles - Level 5 leadership, Transactional and Transformational Leadership, Power and Politics: Definitions of Power, Bases of Power, Power Structure and Blocs, Impression Management, Factors that influence political behavior in organizations. Conflict and Collaboration: Sources of Conflict, Types of Conflict, Process of Conflict, Individual, Interpersonal, group level conflict, Organizational Conflict, Approaches to Conflict Management.

UNIT – IV

Stress Management: Concept of stress, Sources of stress, Effects of stress on humans, Management of Stress. Management of Change: Importance, forces responsible for the change, Resistance to change, overcoming resistance to change, Introduction of change in the organization, and Organizational Development as a tool for introducing change. Organizational Design: Various organizational structures and their effects on human behavior, Organizational climate, Organizational culture, and Organizational effectiveness.

13 **Suggested resource for reading**

1. Robbins, S.P., *India Organizational Behavior*, Prentice-Hall.
2. Davis, K., *Human Behaviour at Work*, Tata Mc Graw Hill
3. Blanchard, H., *Management of Organizational Behaviour*, Prentice Hall India.
4. Ashwatappa, K., *Organizational Behaviour*, Himalaya Publishing House.
5. Borkowski, N., *Organizational Behavior in Health Care*, Jones and Bartlett Learning.
6. Körner M., Ansmann, L., Schwarz B. and Kowalski C., *Organizational Behaviour in Healthcare: Theoretical Approaches, Methods and Empirical Results*, LIT Verlag.
7. Luthans, F., *Organizational Behaviour*, Mc Graw Hill.
8. Pareek, U., *Understanding Organizational Behaviour*, Oxford University.

Note: Latest edition of text books may be used.

Semester III

1	Course Title	Hospital Resource Management
2	**Course Code**	**HAM 301**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) a The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d .Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	This course intends to develop an understanding of the students of concepts, techniques, theories and practices of human resource management in hospitals and about the general information of available drugs related issues applicable in hospital industries.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the importance of human resource management CO2. Use the different methods of recruitment in procuring human resource. CO3. Design relevant appraisal methods for employee compensation, rewards and benefits. CO4. Have basic knowledge about labour laws. CO5. Organize training and development activities to enhance the knowledge, skill and abilities of their workforce. CO6. Apply modern trends in human resource management.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Introduction to HRM, Evolution of Human Resource Management and its role in Hospitals, Organizing the Human Resource department.

UNIT – II

Manpower Planning, Recruitment, Selection and Induction, Training and Management Development, Wage and Salary Administration

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UNIT – III

Employees Performance Appraisal, Working Conditions, Safety, Welfare and Employees Health Services, Promotions, Transfers and Resignations.

UNIT – IV

Collective Bargaining in Hospitals, Discipline and Disciplinary Action, Human Resource Audit, Human Relations in Hospitals, and Teamwork in Hospitals.

13 Suggested Resources for Reading

1. Rao, V. S. P., *Resources Management*, Excel Publishing
2. Flippo, E. B., *Personnel Management,* Tata McGraw Hill
3. Flippo, E. B., *Human Resource Management*, Tata McGraw Hill
4. Sundaresan, M. and Gopalakrishnan, P., *Material Management: An Integrated Approach,* PHI.
5. Raje, V. N., *Drug Store and Business Management*, CBS Publishing and Distributors.
6. Sharma, D. K. and Goyal, R. C., *Hospital Administration and Human Resource Management*, PHI.
7. Fleming Jr., F.L. and McConnell, C.R., *Human Resource Management in Health Care: Principles and Practice*, Jones and Bartlett.
8. Fried, B.J. and Fottler, M.D. *Human Resources in Healthcare*

Note: Latest edition of text books may be used.

1	Course Title	Hospital Planning & Designing
2	**Course Code**	**HAM 302**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) **a.** The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	To provide knowledge of planning, designing and safety management in hospital services.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1.Identify various critical components in planning of the hospital. CO2. Describe the role of hospital environment in delivering quality patient care. CO3.Identify planning and designing considerations while planning for services. CO4. Articulate the functional requirements of the hospital in delivery of patient care. CO5.Role of administrators in the smooth functioning of the hospital.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory ,Interactive sessions ,PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

PLANNING AND ORGANIZATION OF THE HOSPITALS - Roles of the hospital in healthcare-hospital planning and design-outpatient Services the nursing unit-intensive care unit nursing services-effective hospital management directing and leading-controlling – financial management.

UNIT – II

CLINICAL AND ADMINISTRATIVE SERVICES - Radiology and imaging services laboratory services-operation theatre suite pharmacy central sterile supply department hospital infection- materials management-evaluation of hospital services.

UNIT – III

DESIGNING OF HOSPITAL SERVICES - Engineering department - maintenance management- clinical engineering electrical system- air conditioning system- water supply and sanitary system centralized medical gas system- communication system- solid waste management and transportation. Legal Compliance for Building the Hospitals

UNIT- IV

HOSPITAL SUPPORT SERVICES AND SAFETY SYSTEMS - Designing support services and safety management -Admitting department- medical records department- food service department laundry and linen service- housekeeping volunteer department- safety in hospital fire safety- Alarm system- disaster management.

13 Suggested Resources For Reading

1.Kunders, G. D., Hospitals, facilities planning and Management, Tata Mcgraw Hill.
2. Sakharkar, B. M., Principles of hospital administration and planning (2nd Edition), 20 Jaypee Brothers Medical Publishers Pvt Limited.
3.Singh, S., Gupta, S., and Kant, S., Hospital infection control guidelines, principles and practice (1st Edition), Jaypee Brothers Medical Publishers Pvt Limited.
4. Sharma, M., Essentials for Hospital Support Services and Physical Infrastructure, Jaypee Brothers Medical Publishers Pvt Limited

Note: Latest edition of text books may be used.

1	Course Title	Customer- Relationship Management
2	**Course Code**	**HAM 303**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH +40PR+20IA) e. The exam of this course will be of 3hours duration. f. There will be no objective type questions. g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive**.**
8	**Course Objectives**	The main objectives of this course is to enable the students to learn the basics of Customer Relationship Management and Sales Force Automation and Database Marketing.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Interpret the relationship marketing. CO2. Summarize the evolution of customer relationship management. CO3. Apply the sales force automation. CO4. Examine the value chain. CO5. Analyze the customer relationship technologies**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Relationship Marketing - Overview of Relationship Marketing – Basis of Building Relationship – Types of relationship Marketing – Customer Life Cycle. Customer Relationship Management CRM : Overview and Evolution of the Concept – CRM and Relationship Marketing – CRM Strategy – Importance of Customer Divisibility in CRM.

UNIT– II

Sales Force Automation Sales Force Automation – Contact Management – Concept – Enterprise Marketing Management – Core Beliefs – CRM in India.

UNIT– III

Value Chain Value Chain – Concept – Integration Business Management – Benchmarks and Metrics – Culture Change – Alignment with Customer Eco System – Vendor selection.

UNIT-IV

Database Marketing – Prospect Database – Data Warehouse and Data Mining – Analysis of Customer Relationship Technologies – Best Practices in Marketing Technology – Indian scenario**.**

13 Suggested Resources for Reading

1. S. Shajahan, 2006. Relationship Marketing, Tata Mc Graw Hill Education.
2. Paul Green Berg, 2009. Customer Relationship Management. Tata Mc Graw Hill.
3. Philip Kotler and Gary Armstrong. 2017. Principles of Marketing. Pearson Education Pvt Ltd, 17th Edition.

Note: Latest edition of text books may be used.

1	Course Title	Project Work-II and Viva-Voce
2	**Course Code**	**PWII**
3	**Typology of Course**	Project work
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	2 Credits
6	**Max. Marks**	50 Marks
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of the /Project Work Course is to help the student develop his/her ability to apply multi-disciplinary concepts, tools and techniques to solve organizational problems and/or to evolve new/innovative theoretical frame work.
9	**Course Content The Project may take any one of the following forms:** **(i)** Comprehensive case study (covering single organization/multifunctional area problem, formulation analysis and recommendations) **(ii)** Inter-organizational study aimed at inter-organizational comparison/ validation of theory/survey of management services. **(iii)** Evolution of any new conceptual / theoretical framework. **(iv)** Field study (emperical study)	

Semester IV

1	Course Title	Community Health & Management Of National Health Programs
2	**Course Code**	**HAM 401**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Duty Manager- Patient Care Services
8	**Course Objectives**	After completing the community health and management of National health programs module, students will be able to recognize and identify the elements in designing the strategies for healthcare delivery for the community.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. gain understandings on the concepts and dimensions of healthcare services. CO2. get familiarity on current issues in community health. CO3. get in-depth knowledge on the various health promotion activities. CO4. get introduced to the concept of planning and promotion of health education. CO5. get familiarized with Chronic disease management & Consumer health.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	a. The syllabus of this paper has been divided into FOUR units. b. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. c. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. d. All questions carry equal marks.

12 Course Content

UNIT – I

Concepts in Health and Disease : Evolution of medicine, public health and community health, Definition of health: the determinants and relative concept, Agent –Host Environmental factors in health and disease, indices used in the measurement of health, levels of prevention, Epidemiology of common communicable diseases and chronic non-communicable diseases
and condition.

UNIT – II

Health Care Delivery system in India : Introduction, Demography and Family Planning, Maternal and Child Health, Urban Health, Occupational Health, Mental Health, Community Geriatrics, Essential Medicines and Counterfeit Medicines.

UNIT – III

Health Education : Definition, approaches in health education, methods, the barrier to effective communication, principles of health education, International Health Regulations and International classification of diseases, International Health agencies and organizations: WHO, UNICEF, UNOP, World Bank, UNFPA, CARE, IHO, Notifiable diseases 2020.

UNIT--IV

National Health Programs: Objectives, strategy, achievements, critical analysis, Cancer screening and national cancer control program, National AIDS Control Program and NACO, National leprosy eradication program, Universal immunization program, National vector-borne disease control program, RNTCP, RCH, National Program for control of blindness, National Health Policies, National Population Policy, National Rural Health Mission.

13 Suggested Resources for Reading

1. Park, K., *Park's Textbook of Preventive and Social Medicine*, Bhanot Publishers
2. Pagano, M., *Principles of Biostatistics*, Duxbury Press.
3. Arora, P. N. and Malhan, P. K., *Biostatistics*, Himalaya Publishers
4. Barker, D. J. P. and Rose, G., *Epidemiology in Medical Practice*, Churchill Livingstone.
5. Knox, E. G., *Epidemiology in Health Care Planning*, Oxford University Press.
6. Mausner, J. S. and Kramer, S., *Mausner and Bahn Epidemiology-An Introductory Text*, Saunders.
7. Clark, D. and MacMohan, B., *Preventive and Community Medicine*, Little, Brown and Company.

Note: Latest edition of text books may be used.

1	Course Title	Introduction To Epidemiology & Biostatistics
2	**Course Code**	**HAM 402**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Duty Manager-Patient Care Services
8	**Course Objectives**	The course imparts skills and competencies in evaluating screening processes, statistical/epidemiology methods and epidemiological studies.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** apply the knowledge of epidemiology in understanding the healthcare scenario in a population. **CO2.** demonstrate the necessary knowledge, skill and competencies required for good administrator as significant contributor in healthcare. **CO3.** handle various data related to healthcare and help make and implement policies in an organization. **CO4**. recognize and train the workforce to meet the challenges of changing dynamics in healthcare.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

General Epidemiology and Epidemiological Methods: Definitions, concept and evolution of the epidemiological concept, Aims of epidemiology, measurement of morbidity and mortality, epidemiology methods, uses of epidemiology data, association and causation screening, immunity, investigation of an epidemic.

Unit II

Environmental Sanitation and Health aspects: Definition of the environment, Water: waterborne diseases, water pollution methods of purification, water quality, Air-pollution indicator and precaution, radiation: hazards and prevention, lighting, noise, and housing standards. Disposal of wastes and Hospital waste management. Nutrition and Health: Classification and sources of various nutrients, requirements, assessment, nutritional diseases, Genetics and Health: Basic principles, genetic predisposition, prevention and social measure, Disaster Management.

Unit III

Biostatistics: Definition, scope and uses of bio statistics and medical research collection, classification and presentation of statistical data, sources and use of statistics, sampling, numerical summaries (proportions, means, SD, Median and percentile), Graphical summaries.

Unit IV

Health Statistics and Health Information System in India: Methods of registration – fallacies/ difficulties, samples registration schemes, common rates and ratios, normal curve and Poisson distribution, Hypothesis testing, a test of significance, normal test, t-test and chi-square test.

13 Suggested Resources for Reading

1. Park, K., *Park's Textbook of Preventive and Social Medicine*, Bhanot Publishers
2. Pagano, M., *Principles of Biostatistics*, Duxbury Press.
3. Arora, P. N. and Malhan, P. K., *Biostatistics*, Himalaya Publishers
4. Barker, D. J. P. and Rose, G., *Epidemiology in Medical Practice*, Churchill Livingstone.
5. Knox, E. G., *Epidemiology in Health Care Planning*, Oxford University Press.
6. Mausner, J. S. and Kramer, S., *Mausner and Bahn Epidemiology-An Introductory Text*, Saunders.
7. Clark, D. and MacMohan, B., *Preventive and Community Medicine*, Little, Brown and Company**.**

Note: Latest edition of text books may be used.

	Course Title	Hazards And Risk Management
2	**Course Code**	**HAM 403**
3	**Typology of Course**	Core Discipline
14	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40TH+40PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relations Executive**.**
8	**Course Objectives**	**The objective of this course is to enhance risk and disaster management skills in the student with special reference to Hospital Services marketing**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Know the principles of hazards management and peculiarities of their implementation. CO2. Know prerequisites of the safety principles in management of modern organizations. CO3. Be able to use these principles and methods in analyzing and solving problems of organization. CO4. Choose a framework to evaluate the performance excellence of an organization, and determine the set of performance indicators that will align people with the objectives of the organization.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Security Organization and Management: Security Threats and Vulnerabilities of Hospitals, Threat Groups, Sensitive Security Areas, Strategic Security System, Functions of Hospital Security Department, Non-Traditional 'Service' Functions, Security Organization and Physical Security Measures, The Staff Complement, Perimeter Protection System, Implementing Physical Controls, Access Control Concepts, Definition, Means and Components, Need for Security Technology, Security Technology Plan, Computer and Information Security, Selection and Management of Departmental Security Staff/ Contract Security Agency and Security Training, System Choice, Verification of Security Personal, Security Staff Discipline, Effective security Management in Hospitals ,Central Security Control Room, Patrols and Post Procedures / Techniques, Control of Visitors / OPD Patients/ Attendants, Standing Instructions for Security of Cash, Security Budgeting, Security Committee, Periodic Security Audit and Updating of Security Procedures, Bomb Threat / Response.

Unit II

Hospital Acquired Infection (HAI): Objectives, Control and Prevention, Housekeeping Dietary Services, Linen and Laundry, Central Sterile Supply Department (CSSD), Security, Engineering Aspects, Nursing Care, Waste Disposal, Antibiotic Policy, Hospital Infection Control Committee - Composition, Role and Functions, Surveillance, Processing of Information Collected, Mode of transmission, Interruption of Transmission, High-Risk Procedures, Training and Education, Universal Precautions for Health Care Workers.

Unit III

Fire Hazards: Elements of Fire, Fire Hazard Triangle, Causes of Hospital Fires, Fire Protection, Structure Planning and Design Considerations, Buildings: Harness Communication Zone - Building Services, Central Air, Conditioning Facilities, Electric 29 Installations, Special Hazards, Fixed Installations, Hazards Associated with Furnishing Material,Curtains Upholstery, Dresses, Bed and Bedding Materials, Water Supply, Fire Points and Escape Routes, Fuel Store, Manual Call Points, Means of Escape and Evacuation, Risk Evaluation.

Unit IV

Disaster Management: Objectives, Basic Concepts, Disaster Classification, Disaster Process, Spectrum of Disaster Management, Special Characteristics, Principles of Disaster Planning, Disaster and Health Problems, Organization for Medical Relief, Principles of Mass Casualty Management, Objectives of and Need for Hospital Disaster Plan, Disaster Committee, Organization, Role and Responsibilities, Organizing Disaster Facilities, Disaster Response, Alert and Recall, Deployment, Disaster Administration, Disaster Manual, Disaster Drill. Disaster Management Act 2005: National Authority, National Executive Committee, State Disaster Management Authority, District Disaster Management Authority, NDRF, Disaster Management Guidelines by WHO.

13 **Suggested Resources for Reading**

1. Singh, S. K., *Safety and Risk Management*, Mittal Publishers.
2. Diwan, J. H., *Safety, Security and Risk Management*, APH.
3. Ayers, S., and Garmvik, A., *Text Book of Critical Care*, Holbook and Shoemaker.

Note: Latest edition of text books may be used.

1	Course Title	Financial Management
2	**Course Code**	**HAM 404**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40TH+40PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive.
8	**Course Objectives**	**The main objectives of this course are to:** 1. Guide the students to develop a framework for understanding financial and managerial reports. 2. Train on financial decisions for mangers who are working in health care institutions**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Explain the general objectives of financial management and apply the fundamental principles, concepts and techniques of financial management CO2. Construct the financial statement and analyze and compare the same. K6 CO3. Develop fund flow, cash flow statements and computing of operation problems. CO4 . Assess the suitability of different financing options for organizing and managing given project. CO5. Compare the financing costs and benefits of various courses of action and provide output by utilizing appropriate appraisal techniques**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Financial Accounting : Definition – Accounting Principles – Concepts and Conventions – Trial Balance – Final Accounts (Problems) – Depreciation Methods - Straight Line Method, Written Down Value Method, Sinking Fund Method. Financial Statement Analysis : Objectives – Reorganizing the Financial Statement Information – Techniques of Financial Statement Analysis: Comparative Statements, Common – Size Statement, Trend Percentage – Accounting Ratios: Simple Problems – Dupont Analysis.

UNIT– II

Fund Flow Statement : Statement of Changes in Working Capital – Computation of Fund From Operations – Workings for Computation of Various Sources and Uses – Preparation of Fund Flow Statement – Cash Flow Statement Analysis – Computation of Cash From Operations Problems – Distinction Between Fund Flow and Cash Flow Statement. Problems.

UNIT– III

Cost Accounting : Meaning – Distinction Between Financial Accounting and Cost Accounting – Cost Terminology: Cost, Cost Centre, Cost Unit – Elements of Cost – Cost Sheet – Problems. Budget, Budgeting, and Budgeting Control – Types of Budgets – Preparation of Flexible And Fixed Budgets, Master Budget and Cash Budget – Problems – Zero Base Budgeting.

UNIT-IV

Marginal Costing : Definition – Distinction Between Marginal Costing and Absorption Costing – Breakeven Point Analysis – Contribution, P/V Ratio, Margin of Safety – Decision Making Under Marginal Costing System – Key Factor Analysis, Make or Buy Decisions, Export Decision, Sales Mix**.**

13 Suggested Resources for Reading

1. R L Gupta. M Radhaswam, 2015. Advanced Accountancy: Theory, Method and Application - Vol. 1, Sultan Chand & Sons First edition.
2. Khan & Jain ,2010. Management Accounting, Tata McGraw - Hill Education, 5th Edition.
3. S K Maheshwari S N Maheshwari, 2018. A Textbook of Accounting for Management, Vikas Publishing House Pvt Ltd ,4th Edition.

Note: Latest edition of text books may be used.

SEMESTER V

1	Course Title	Medical Ethics And Legal Issues In Hospital
2	**Course Code**	**HAM 501**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator**.**
8	**Course Objectives**	To familiarize the students with legal issues involved in hospital management.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. apply the knowledge of ethics in the functioning of the hospital. CO2. handle various issues related to healthcare setup and also manage the hospital with the various issues that can arise from the legal perspective. CO3. recognize and train the workforce to meet the challenges of changing dynamics in healthcare scenario in terms of the regulations that govern the operational aspects of the hospital CO4. demonstrate the necessary knowledge, skill and competencies required for good administrator as significant contributor in healthcare
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory ,Interactive sessions ,PPT, Practical and Group Discussions ,Case Studies etc.
11	**Instructions for paper setters and candidates**	**1.** The syllabus of this paper has been divided into FOUR units. **2.** Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. **3.** The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

Unit I

Medical Ethics : Introduction to Legal aspects in health care: health as a fundamental right, Hospital services and law: Medical ethics including Hippocratic Oath, declaration of Geneva and International Code of medical ethics and violation of these ethics, informed consent.

Unit II

Rights and Responsibilities of Patients : Rights and responsibilities of patients: confidentiality issues with respect to contractual obligations towards the patients, electronic medical records and Health. Liability and accountability towards patients and third parties, and employees (vicarious liability).

Unit III

Medical Negligence : Medical negligence -definition, The Bolam test, Civil and criminal negligence, relevant sections of Indian Penal code, defence against medical negligence, The Consumer Protection Act-Ethics Introduction, its scope, various levels of consumer courts (District Forum, State Commission, National Consumer Dispute Redressal Forum) and procedure followed in each, Professional Indemnity, Legal Implications in Medical Emergencies, Euthanasia- its types and the international and national Perspective, the concept of Living Will- Medical ethics.

Unit IV

Health Care Laws : A brief account of various laws related to healthcare: -Human Organs Transplantation Act, Inquest and its proceedings, The Drugs and Magic Remedies Act, Registration of Birth and Death Act, PNDT Act (1994) and MTP Act (1971). Regulatory framework of hospitals – The Indian Medical Council Act, Regulatory requirements to set up a Blood Bank, Legal aspects of Assisted Reproductive Techniques.

13 Suggested Resources for Reading

1. Francs, C.M., *Medical Ethics*, Jaypee Publications.
2. Pozgar, G. D., *Legal Aspects of Health Care Administration*, Jones and Bartlett Publications.
3. Malik, V., *Laws Relating to Drugs and Cosmetics*, Eastern Book Co.
4. Saundbar, R., *Medical Ethics: A Guide to Professional Conduct*, Forgotten Books.
5. Shenoy, P. D., *Medical Negligence: What Doctors, Patients and Hospitals Should Know*, Sterling Publications.
6. Koley, T. K., *Medical Negligence and the Law in India*: *Duties, Responsibilities, Rights*, OUP India.

Note: Latest edition of text books may be used.

1	Course Title	Hospital Materials Management
2	**Course Code**	**HAM 502**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The purpose of this course is to learn the specific methods of materials planning, purchasing, processing, stocking and dispensing, including equipment purchasing and maintenance.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. gain insights on concepts of material management in hospitals. CO2. get knowledge on various objectives and elements of purchasing. CO3. get in-depth knowledge on the purchase of equipment's and maintenance. CO4. get introduced to various techniques of distribution channels and functions. CO5. get familiarized with inventory management and control.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12 Course Content

Unit I

Materials Management : Introduction, Definition and Function, Goals and Objectives of Materials Management, Materials Cycle, Functions of Materials Manager, Problems and Issues in Hospitals, Information Systems for Materials Management.

Unit II

Purchasing : Objectives and Elements of Purchasing, Purchasing System, Purchasing Cycle, Purchase Procedures, Legal and Ethical Aspects, Conditions of Contract, Financial Rules, Arbitration, Equipment Purchase and Maintenance - Planning and Selection of Equipment, Import of Equipment, Equipment Utilization and Operation, Equipment Repair and Maintenance, Equipment Audit.

Unit III

Inspection, Storage and Distribution of Materials: Planning Consideration of Stores, Inspection and Verification of Materials, Storage of Materials, Distribution of Materials, Condemnation and Disposal.

Unit IV

Scientific Inventory Management : Codification and Standardization, value Analysis, Inventory Control, Lead Time, Safety Stock and Reorder Level, Economic Order Quantity (EOQ), Selective Controls, and Case Studies on Inventory Control.

13 Suggested Resources for Reading

1. Gupta, S. and Kant, S., *Hospital Stores Management: An Integrated Approach,* Jaypee Publications. 33
2. Gopalakrishnan, P. and Sundaresan, M., *Materials Management: An Integrated Approach,* Prentice-Hall of India Pvt. Ltd.
3. Housley, C. E., *Hospital Material Management*, Lippincott Williams and Wilkins
4. Sanderson, E. D., *Effective Hospital Material Management,* Aspen Publications.

Note: Latest edition of text books may be used.

1	Course Title	Market Research And Consumer Behavior In Healthcare
2	**Course Code**	**HAM 503**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination. Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Relation Manager
8	**Course Objectives**	The course is intended to help students get acquainted with various research concepts, techniques and procedures and to develop their ability to conduct, evaluate, use and Present research findings with special reference to Health Services. The course will also help students to understand the various factors affecting consumer behavior and to understand the process of consumers buying Health Services.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Explain and apply the key terms, definitions, and concepts used in the study of consumer behavior. CO2. Demonstrate how as a marketer you can use your knowledge of consumer behavior concepts to develop better marketing programs and strategies to influence those behavior. CO3. Critically evaluate the effectiveness of various advertisement and promotions and their attempts to influence the behavior of individuals. CO4. Complete a project that demonstrates both your working knowledge and analytical skills in assessing the consumer decision-making process. CO5. Analyze the trends in consumer behavior, and apply them to the marketing of an actual product or service.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Introduction to Consumer Behaviour: Introduction, Applications of consumer behaviour, Consumer behaviour and marketing strategy, Situational characteristics and consumption behaviour, Consumer decision-making process, and post-purchase behaviour.

Unit II

Factors Influencing Consumer Behaviour: Consumer-specific factors influencing buying behaviour, psychological processes (motivation, perception, learning and memory) affecting consumer behaviour, the buying motive, high involvement and low involvement buying situations, Consumer behaviour in adapting new products, participants in buying decisions. Group influence- Definition, types of the group (primary, secondary, formal etc.), Family influence, roles, FLC application.

Unit III

Introduction to Marketing Research: Meaning, scope and importance of marketing research, Defining the market research problem and developing an approach, Research design formulation, Rating Scales: Juster, Likert, Semantic Differential, Thurston, Attitude Scales. Preparing a written research report, organization of the report, and Ethical issues in marketing research.

Unit IV

Consumer Behaviour for Hospital Services: Case studies on Consumer behaviour for Hospital services. Applications of marketing research in the Hospital industry: Research 35 design classification, Product research-developing products, Specifications and attributes, Concept of test marketing. Advertising research, Market and sale analysis research, Sales forecasting, and Demand measurement.

13 Suggested Resources for Reading

1. Gupta, S.L., *Marketing Research*, Excel Books
2. Beri, G.C., *Marketing Research*, McGraw Hill.
3. Bradley, N., *Marketing Research - Tools and Techniques*, Oxford University Press.
4. Bryman, A. and Bell, E., *Business Research Methods*, Oxford University Press.
5. Churchill, G. A. and Iacobucci, D., *Marketing Research,* Dryden Press

Note: Latest edition of text books may be used.

1	Course Title	Strategic Management In Healthcare
2	**Course Code**	**HAM 504**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	This subject will integrate all management concepts to look at the organization from futuristic - both long and short term point of view. The emphasis will be on organizational analysis, planning and formulation of strategies.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Will be able to get the knowledge on the concept of strategic management and its process. CO2. Will be able to get familiarity on environmental analysis and its techniques. CO3. Will be able to get in-depth knowledge on the various types of strategies and its formulation. CO4. Get acquaint with strategy implementation and evaluation. CO5 . Get introduced to different strategic management practices in healthcare.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Introduction to strategic management Organizational mission: Philosophy, policy, vision, values. Strategic management process, nature of strategic decisions, organizational Strategies, strategic business units. Strategic objectives, generic corporate strategy alternatives, porter's value chains concept and applications.

UNIT– II

Industry characteristics analysis: Nature and degree of competition, buyer power, supplier power, entry barriers, competitive power of substitutes

UNIT– III

Organizational analysis: Strength and weakness analysis, identification of opportunities and threats, critical environmental variable, critical success factor, environmental Scanning.

UNIT-IV

Corporate strategy analysis BCG growth share matrix, GE business portfolio matrix. Introduction to Balance Score card. Strategies for low market share firms, for dominant firms, for firms in stagnant industries, contingency planning. Functional strategy analysis, internal and external considerations, product life cycle. Implementation: Corporate culture, structure, human resource, rewards. Control of strategies, strategic diagnosis, operational diagnosis.

13 Suggested Resources for Reading

1. Oster Sharon M. Strategic Management for non profit organizations, Oxford publishers, New Delhi.
2.Lawrence R. Jauch and WF Glueck, Business Policy and strategic management, 6th edition, Frank brothers, New Delhi, 2003.
3.Srivastava; Management Policy ,And Strategic Management; Himalaya Publications, Mumbai. . Allio, Robert J; The practical Strategist - Business and corporate strategy for 1990s, Indus publications, 1994.
4.Kazhmi, Business Policy and strategic management, Tata Mc Graw Hill, New Delhi, 2002.

Note: Latest edition of text books may be used.

1	Course Title	Patient Care & Healthcare Technology
2	**Course Code**	**HAM 505**
3	**Typology of Course**	Core Discipline
14	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main objective of the course is to provide the insights on the basic and advance concepts in patient care and healthcare technology.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Will be able to gain insights on patient care rights and regulations. CO2. Will be able to get knowledge on types of policies and procedures of hospitals. CO3 .Will be able to get in-depth knowledge on the concept of patient care. CO4 .Get introduced to the concept of healthcare technology and related issues. CO5 Get introduced to planning and evaluation methods of healthcare technology.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Introduction: Patient Rights – Patient Behaviour – Models of Patient Behaviour –Patient Motivation – Patient Perception – Attitudes – Attitude Change – Personality, Patient Involvement and Decision Making, Reference Group Influence – Opinion Leadership – Family Decision Making.

UNIT – II

Policies and Procedures of the Hospitals for patients and personnel: Service Buying Behaviour – Psychographics – Lifestyles – Information Search Process – Evaluating Criteria Audit of Patient Behaviour.

UNIT – III

Patient Care: Introduction, Importance of improving the quality care of patients, role of natural and human resources in patient care management, patient counseling: for surgical procedures, for treatment, grief counseling; protocols, Medicare standards, Role of Medical Superintendent, Hospital Administrator, Resident Medical Officer, Night Duty Executive; Public and guest relation; importance in patient care.

UNIT – IV

Concepts And Issues Related to Healthcare Technology: Introduction; Problems and constraints associated with healthcare Technology; Present trend in Healthcare Technology; Hospitals and Technology; Dealing with Technological Problems. Planning Process For Introduction Of Technology In Healthcare: Healthcare Technology in developing countries; Planning and adopting appropriate Technology in healthcare; Mechanism to ensure appropriate use of healthcare Technologies; Developing sources of information on hospital technology; Evaluation methods of health technology; Application of technology: In diagnostic Service areas (Radiology, Lab Services etc), In clinical Services areas (Nephrology, Urology, Cardiology etc), In therapeutic services, and In patient support areas.

13 Suggested Resources for Reading

1. Srinivasan, A.V. (ed), Managing a Modern Hospitals, Response Books, New Delhi, 2000
2. Wiley Blackwell, Improving Patient care BMJI Books.

Note: Latest edition of text books may be used.

1	Course Title	Brand Management
2	**Course Code**	**HAM 506**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	**The main objectives of this course are to:** 1. Enable students to understand the basic concepts of Branding and Strategic Brand Management. 2. Learn Brand Promotion, Brand Extension and Brand Equity.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Explain branding and it's functions. CO2. Build a strong brand and position in the competitive market. CO3. Construct the brand promotion methods. CO4. Analyze the different type of brand extension. CO5. Discover the brand equity management**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions ,Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT I

Basic Understanding of Brand : Definition - Branding Concepts – Functions of Brand - Significance of Brands – Different Types of Brands – Co branding – Store Brands.

UNIT II

Strategic Brand Management Strategic Brand Management Process : Building a Strong Brand – Brand Positioning – Establishing Brand Values – Brand Vision – Brand Elements – Branding for Global Markets – Competing with Foreign Brands.

UNIT III

Brand Promotion Brand Image Building : Brand Loyalty Programmes – Brand Promotion Methods – Role of Brand Ambassadors - Celebrities – On line Brand Promotions. Brand Extension Brand Adoption Practices : Different Type of Brand Extension – Factors Influencing Decisions for Brand Extension – Re-Branding and Re-Launching – Brand Rejuvenation.

UNIT IV

Brand Equity Branding Strategies : Measuring Brand Performance – Brand Equity Management - Brand Equity Measurement – Brand Leverage - Global Brand Audit – Role of Brand Managers – Branding Challenges & Opportunities – Case Studies.

13 Suggested Resources for Reading

1. Kevin Lane Keller, Vanitha Swaminathan, 2019. Strategic Brand Management: Building, Measuring and Managing Brand Equity, Prentice Hall, 5th Edition.
2. Harsh V.Verma, 2006 . Brand Management : Text & cases, Excel Books India.
3. Paul Tmeporal, 2000. Branding in Asia, John Willy & sons (P) Ltd, New York
4. Ramesh Kumar, 2002.Managing Indian Brands, Vikas Publication, India.

Note: Latest edition of text books may be used.

1	Course Title	Project Work-III and Viva-Voce
2	**Course Code**	**PW1II**
3	**Typology of Course**	Project work
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	2 Credits
6	**Max. Marks**	50 Marks
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of the /Project Work Course is to help the student develop his/her ability to apply multi-disciplinary concepts, tools and techniques to solve organizational problems and/or to evolve new/innovative theoretical frame work.
9	**Course Content The Project may take any one of the following forms:** **(i)** Comprehensive case study (covering single organization/multifunctional area problem, formulation analysis and recommendations) **(ii)** Inter-organizational study aimed at inter-organizational comparison/ validation of theory/survey of management services. **(iii)** Evolution of any new conceptual / theoretical framework. **(iv)** Field study (emperical study)	

SEMESTER VI

1	Course Title	Hospital Administration
2	**Course Code**	**HAM 601**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of the course is to equip the learners with advance concept of hospital administration and hospital management services**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Will be able to gain insights with the concept of hospital and its classification. CO2. Will be able to get knowledge on hospital administration and its functions. CO3. Will be able to get in-depth knowledge on the various types of services provided by the hospitals. CO4. Get introduced to legal framework of hospital management structure. CO5. Get introduced to recent advances in hospital administration.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Concept of Hospitals: Definition, philosophy and objectives of Hospital; Classification of Hospitals; Hospital as a System and its peculiarities; Intramural and Extramural Functions of a Hospital; Managerial activities in a Hospital; Relationship between a Hospital and its community.

UNIT – II

Introduction to Hospital Administration: Meaning, nature and principles of Administration; Administration vs Management; Meaning and rationale of Hospital Administration; Roles of Hospital Administration; Skills of Hospital Administration; Types of Hospital Administrators; Professional bodies of Hospital Administrators; Code of Ethics for Hospital Administrators.

UNIT – III

Overview of Hospital Services: Administrative Services; Medical and Ancillary Services; Nursing Services; Supportive Services: Pharmacy, medical stores, housekeeping, ward management, CSSD, Laundry, Dietary, Security, and Transport.

UNIT – IV

Hospital Management: Levels and Roles: Governing Board; Executive Board and Advisory Board; CEO; Medical Administration Nursing Administration; Hospital Administration; Middle Level Mangers in Hospital and their Responsibilities; Structuring Hospital Organization. Evaluation of Hospital Services; Management Techniques in Hospitals; Recent Advances in Hospital Administration.

13 Suggested Resources for Reading

1. Sakharkar, B. M., &Jaypee Brothers (Jaypee digital). (2009). Principles of Hospital Administration & Planning. (Jaype eBooks.) Jaypee Brothers Medical Publisher (P) Ltd
2. Srinivasan, S. (1982). Management process in health care. New Delhi: Voluntary Health Association of India.
3. Joshi DC, Joshi, Mamta, (2009). Hospital administration. Jaypee Brothers Medical Publications
4. Gupta, M. C., &Mahajan, B. K. (2003). Textbook of preventive and social medicine. New Delhi: Jaypee Brothers Medical Publishers.

Note: Latest edition of text books may be used.

1	Course Title	Nursing Administration and Management
2	**Course Code**	**HAM 602**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main objective of the course is to familiarize the students with the role of nursing and its administration existing in the hospitals and its growth.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. develop managerial and administrative skill in Nursing. CO2. gain knowledge on Functions & Principle of Nursing Administration. CO3. understand Clinical Care Delivery and Optimal Patient outcomes. CO4. develop skill on planning & organization of Nursing services. CO5. understand Objective & models of Clinical leadership. CO6. gain knowledge on Nursing Quality & Total Quality Management. CO7. gain knowledge on effective communication in nursing. C08. understand financial management in Nursing. CO9 .understand introduction and factors influencing latest trend in nursing.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Nursing Administration : Nature of nursing, nursing as a profession, Code of ethics, Functions of Professional Nursing, Ethical and Legal obligations of Nurses. Need for Nursing Administration. Nursing Implication of the Patients' Rights, Nursing Implication for Changes in Healthcare delivery system.

UNIT – II

Planning For hospital nursing services, Philosophy and Objectives of nursing services. Nursing services and Computers. Nursing Service Department, Organization and Management of the Nursing Service Unit.

UNIT – III

Nursing Management : Ward Management, Equipment Management, Modes of organizing patient care or Nursing Care Delivery System, Time Management, Nursing Supervision. Application of Administration/Management Concepts at Different Levels in Hospitals, Role of Chief of Nursing, the Role of Supervisor. Evaluation, Accountability in Nursing

UNIT- IV

Discipline in Nursing, Healthcare Economics and Nursing, Clinical Supervision, and Quality Control: Nursing Audit, Nursing Standards, Nursing Care Standards in Ward Management, Patient Care Audit, Evaluating the quality of nursing Care. Peer Review, Quality Circles, Risk Management. Staff Development: Types of Staff Development, Continuing Education, InService Education, Staff Development Process, Training Methods and Techniques.

13 Suggested Resources for Reading

1. Srinivasan, A.V., Managing a Modern Hospital, Response Books.
2. Kumar, A., Encylopedia of Hospital Administration and Development, Anmol Publications.
3. Park, K., Textbook on Hygiene and Preventive Medicine, Banarsidas Bhanot Publishers.
4. Neeraja, K.P., Textbook of Communication and Education Technology for Nurses, Jaypee Brothers.
5. Moglli, G.D., Paramedics 6-in-1 Handbook, Jaypee Brothers.
6. Basavanthappa, B T., Nursing Administration, J P Brothers Medical Publishers
7. Raman, K. V., Hospital Management: Text and Cases, Pearson.

Note: Latest edition of text books may be used.

1	Course Title	Biological Waste Management
2	**Course Code**	**HAM 603**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The Objective of the Course is to familiarize the learner with the importance, techniques and procedures involved in the management of Hospital Waste**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. gain insights on different types of biomedical wastes. CO2. get knowledge on the need for disposal of biomedical waste. CO3. get in-depth knowledge on various types of health hazards occurred due to biomedical waste. CO4. get introduced to various legislation and policies on healthcare waste management. CO5. gain knowledge on international management of hospital waste in developing countries..
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Hospital Hazards: Meaning, Types, Physical–Biological Mechanical, Psychological, Its Impact on Employees, Preventive Measures. Hospital Hazards Management: Meaning, Need, Principles, Purpose.

Unit II

Hospital Hazards: Meaning, Types, Physical–Biological Mechanical, Psychological, Its Impact on Employees, Preventive Measures. Hospital Hazards Management: Meaning, Need, Principles, Purpose.

Unit III

Biomedical Waste Management: Meaning, Categories of Biomedical wastes, Disposal of biomedical waste products, Incineration and its Importance, Standards for Waste Autoclaving, Micro Waving, and Deep Burial, Segregation, Packaging, Transportation, Storage, Guidelines for Management of Healthcare Waste as per biomedical waste management rules 2016- Biomedical Waste Management at outreach activities and by occasional generators - Biomedical Waste Management (Amendment Rules) 2018, Revised guidelines for handing Biomedical Waste during pandemics.

Unit IV

Human Waste Disposal and Sewage Disposal: Diseases carried from excreta, Sanitation barrier, Methods of Excreta disposal, Sewage wastes: Meaning, Composition, Aims of Sewage disposal, Decomposition of Organic Matter, Modern Sewage Treatment, Drawbacks of improper disposal of wastes – Solid and liquid.

13 Suggested Resources for Reading

1. Basakar, S., *Hospital Waste Management,* Jaypee Brothers Publishers.
2. Acharaya, D. B., *Practical Handbook on Hospital Waste Management*, Jaypee Brothers Publications.
3. Sharma, M., *Hospital Waste Management and its Monitoring*, Jaypee Brothers Publishers.
4. Saxena, A., *Hospital Waste Management: Principles and Practices*, Navyug Books International**.**

Note: Latest edition of text books may be used.

1	Course Title	Hospital Operations Management
2	**Course Code**	**HAM 604**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The course aims to equip the learners with decision-making skills in hospital planning, scheduling and control of production and operation functions**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the role and functions of operations within the organization. CO2. Analyzing the key concepts, theories and techniques for the adequate functioning of the hospital. CO3. Identify and demonstrate the dynamic nature of the environment in which the organization is functioning and the implications of it the competitive market. CO4. Applying the conceptual frameworks, theory and techniques to various inventory management aspects in the hospital**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Introduction to Management: Approach, Planning, Decision Making, Organizing, Staffing, Job Designing, Motivating, Leadership, Communication, Materials Management, Financial Management, and Industrial Relations in Hospitals.

UNIT – II

Hospital Planning: Aims, Factors Affecting, Hospital Planning Process, Hospital Architect, Zoning, Design, Hospital Project, Construction.

UNIT – III

Support and Utility Services in Hospitals: MRD, CSSD, Laundry, Housekeeping, Transportation, Equipment Management, Public Relations, Dietary Services, Mortuary.

UNIT – IV

Risk Management in Hospitals: Hospital Waste Management, Disaster Management, Hospital Security Services, Occupational Safety in Hospitals, Nosocomial Infections.

13 Suggested Resources for Reading

1. Sharma, M., Essentials for Hospital Support Services and Physical Infrastructure, Jaypee Brothers Publications.
2. Joshi, D.C., Hospital Administration, Jaypee Brothers.
3. Kunders, G.D., Hospitals Planning and Management, Tata Mc Graw Hill
4. Dasgupta, J., Hospital Administration and Management: A Comprehensive Guide, Jaypee Publications.
5. Gupta, S. K., Kant. S. and Chandrashekhar, R., Modern Trends in Planning and Designing of Hospitals: Principals and Practices, Jaypee Publications.
6. Goel, S.L. and Kumar, R., Hospital Administration and Management: Theory and Practice, Deep and Deep Publications.

Note: Latest edition of text books may be used.

1	Course Title	Medical Terminology And Procedures
2	**Course Code**	**HAM 605**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main objective of this course is to familiarize the students with medical jargon and the human body system to be an effective member of operational team.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to:** CO1. Identify and define combining forms, prefixes and suffixes pertaining to the Human anatomy and physiology system. CO2. Correctly spell and pronounce medical terminology associated with the Human anatomy and physiology system. CO3. Identify and define selected anatomical and pathology terms associated with the Human anatomy and physiology system. CO4. State and describe diagnostic procedures, treatments and therapeutic procedures associated with the Human anatomy and physiology system. CO5. State abbreviations associated with the Human anatomy and physiology system.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory ,Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Fundamentals of Medical Terminology : Word Roots- Prefix – Suffix - Abbreviations & Symbols. Introduction to Anatomy & Physiology. Organs & Systems Gastro Intestinal – Respiratory – Circulatory – Renal – Reproductive – Nervous. Common Diseases & Procedures Gastro Intestinal – Cholecystitis – Cholelithiasis – Appendicitis - Intestinal Obstruction – Hernia – Peritonitis.

UNIT– II

Common Diseases & Procedures Respiratory – Tuberculosis - Bronchial Asthma - Respiratory Failure - Pulmonary Embolison – Pneumonia -Branchosopy, Pulmonary Function Test, Cardio-Pulmonary Resuscitation 3 hours Module 6 Circulatory Hypertension - Coronary Artery Disease – Arrhythmias - Cardiac Arrest.

UNIT– III

Renal Nephrotic Syndrome - Urinary Tract Infection - Renal Failure - Renal / Bladder Stones - Intravenous Pylography, CystoscopyNervous Stroke (Cerebro Vascular Accident) - Brain Tumor - Brain Injuries - Spinal Cord Injuries - Lumbar Puncture, Myelography, CT Scan, MRI, EEG, EMG.

UNIT-IV

Reproductive Female – Breast Cancer/Self Examination - Menstrual Disorders, Dysmenorrhoea - Premenstrual Syndrome (PMS) -Menorrhagia Ovarian Cyst, Fibroids - Malignancy, Infertility Mammography, Ultra Sound, Laparoscopy, IVF, Tubectomy, D & C. - Male - Prostate Enlargement, Hydrocele, Impotence.

13 Suggested Resources for Reading

1. Gerard J. Tortora and Bryan Derrickson, 2016. Principles of Anatomy & Physiology, Wiley & Sons, 15 th Edition.
2.Anne Waugh and Allison Grant, 2014 Anatomy & Physiology in Health and Illness Churchil Livingstone, 12th Edition.
3. Evelyn C Pearce, Anatomy & Physiology for Nurses , Jaypee Brothers, New Delhi, 16th Edition.
4. Dorland’s Pocket Medical Dictionary.
5. Taber’s Cyclopedic Medical Dictionary , F.A.Davis PT collection, 23rd Edition
6. Sampath Madhyastha ,2015. Manipal Manual of Anatomy , CBS Publication, 2nd Edition

Note: Latest edition of text books may be used.

1	Course Title	Research Methodology
2	**Course Code**	**HAM 606**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	At the end of this course, the students will be able to design research methods, collect and analyze data, prepare research reports and independently conduct research in their field of study/area of work.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. build their own formulations, to expand existing formulations, to critically evaluate the impact of model assumptions and to choose an appropriate solution technique for a given formulation. CO2. know why organizational research is undertaken, and the audiences that profits from research studies. CO3. identify the overall process of designing a research study from its inception to its report.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks**.**

12 Course Content

Unit I

Research Methodology: Nature and Scope of Research Methodology. Importance of Research in Important Areas.

Unit II

Types of Research: Exploratory Research, Prescriptive Research, Experimental Research.

Unit III

Methods of Data Collection: Primary Data-Secondary data. Designing of a Questionnaire, Sampling - Probability sampling and non-probability sampling methods.

Unit IV

Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals, Case Studies.

13 Suggested Resources for Reading

1. Kothari, C. R., *Research Methodology Methods and Techniques,* Wiley Eastern Ltd.
2. Rao, S. P. S. S. and Richard, J., *An Introduction to Biostatistics – A Manual for Students in Health Sciences,* Prentice – Hall of India Pvt. Ltd.
3. Polgar, S. and Thomas, S. S., *Introduction to Research in the Health Sciences,* Churchill Livingstone Publications
4. Mahajan, B. K., *Methods in Biostatistics,* Jaypee Brothers Medical Publishers P. Ltd.
5. Gupta, S. P. and Gupta, M. P., *Business Statistics,* Sultan Chand and Sons**.**

Note: Latest edition of text books may be used.

1	Course Title	Management & Organization Of Medical Records
2	**Course Code**	**HAM 607**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	This course aims to develop awareness among students regarding the concept of location, layout and infrastructure and staff requirements. They are expected to develop general procedures and policies to manage super speciality services.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Outline the history and role of medical records in health care sector and to interpret its legal aspects. CO2. Organize the types of medical records quality assurance aspects in health care industry. CO3. Examine the security of medical records and documentation practices in hospital administration. CO4. Interpret to the documentation issues during treatment of care. CO5. Build an effective electronic medical record systems to upgrade with the health care organization.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Medical Record Organization and Procedures: Role of medical records in healthcare delivery, General medical records standards and policies, legal aspects of medical records and electronic health records.

Unit II

Medical Record Procedures: Medical Audit, International Classification of diseases -10, Organization and Management of medical record department.

Unit III

Medical Record Procedures II: Central registration and appointment section, outpatient clinic, Accident and emergency service, admissions office, ward, processing of outpatient and inpatient records, Hospital statistics, filling and retrieving of patient files, medical record library journal.

Unit IV

Development and Implementation of Electronic Health Records: Challenges in hospital IT and networking design, Tips for evaluation of electronic health record software, Roadmap for successful implementation of HER, Amalgamation of Manual Records with Electronic Health Records, Implementation of the personal health record.

13 Suggested Resources for Reading

1. Sakharkar, B. M., *Principles of Hospital Administration and Planning*, Jay Pee Publications.39
2. Mogli, G. D., *Medical Records Organization and Management*, Jay Pee Publications.
3. Dasgupta, J., *Hospital Administration and Management: A Comprehensive Guide*, Jay Pee Publications
4. Gupta, S. K., Kant, S. and Chandrashekhar, R., *Modern Trends in Planning and Designing of Hospitals-Principles and Practice*, Jay Pee Publications.
5. Mogli, G. D., *Health Records Paper to Paperless,* Jay Pee Publications.
6. Gupta, S. and Kant, S., *Hospital and Healthcare Administration*, Jaypee Publications.
7. Farmer, J. C., *Hospital Emergency Management Guidebook*, JCI Resource.

Note: Latest edition of text books may be used.

SEMESTER VII

1	Course Title	Healthcare Economics
2	**Course Code**	**HAM 701**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The primary aim of this subject is to provide a clear, concise introduction to micro economic concepts, health economics as applied to hospital and health care sector**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1 gain insights on health economics and its application to business decisions. CO2 get knowledge on the resource allocation and its classification. CO3 get in-depth knowledge on the market structure and price output decisions. CO4 get introduced to various budgets and its allocation in medical field. CO5 get introduced to medical tourism and government policy on medical tourism.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Nature and scope of Economics: Basic Economic Concepts - Circular Flow of Economic Activity - Using Economics to Study Health Issues - Nature and Relevance of Economics to Health and Medical care, Concept of Health- Health Determinants, Valuation &-Measurement of Health Demand analysis. Indifference curve analysis. Production possibility frontier.

UNIT– II

Utility Analysis: Laws of Demand and Supply, The Demand for Health, The Demand for Medical Care, Concept of Elasticity, Production, Cost functions, Policies to Contain Costs Supply and cost analysis. Demand and Supply:
Market forms- Demand and Supply of Medical and Healthcare Services, Price determination under various configuration.

UNIT– III

Analyzing Medical care Markets :The Medical Care Market Place, The competitive Market Model, Market Failure in Medical Markets, Government Intervention in Medical Markets. Market for Healthcare Professionals Application of the Theory of Labor Markets in the case of Health Care Professionals, The Market for Physicians Services, Models of Physician Behavior, The Market for Nursing Services, The Market for Dental Services. Market for Hospital services Alternative Models of Hospital Behavior-Utility Maximizing Model, Physician Control Models, The Trend towards Multi Hospital Systems.

UNIT-IV

Healthcare system: Indian Healthcare system - Health Policies - Expenditure and Allocations Under Five-Year Plans-Role of Private Sector and PP.

13 Suggested Resources for Reading

1.Dwivedi, D.N., Microeconomic Theory, Vikas Publications, New Delhi, 1996
2.James Henderson, Health Economics and Policy, South - Western College Publishing, 18 International Thomson Publishing, USA, 1999
3.Rexford E. Santerre, Stephen P. Neun, Health Economics, Dry den Publishers, Florida (U.S.A.), 2000
4.Kenneth Black, Jr., Harold D.Skipper,Jr., Life & Health Insurance, 13th edition, Pearson Education, New Delhi, 2003
5.Insurance Chronicle, ICF Al Publications, Hyderabad .
6.The Insurance Times, Kolkata Insurance Institute of India publications

Note: Latest edition of text books may be used.

1	Course Title	Healthcare Insurance
2	**Course Code**	**HAM 702**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main objective of the course is to familiarize the students with the concept of health insurance, its scope and applicability.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Identify the process of insurance management and claim settlement procedures. CO2. Understand various risks involved in insurance management. CO3. Be competent in understanding the marketing mechanisms of health insurance. CO4. Analyze the requirements of accreditation process.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Health Insurance for Individuals and Groups: Standard/Individual, Floater, Critical Illness, Cancer Insurance, Group Health Insurance, Cradle care insurance, Road Safety Insurance. Coverages , Exclusions, limitations, terms and conditions, Role of Third-Party Administrator for settlement of the claim.

Unit II

Health Economics and Financing: Health Economics- Financing models across world and in India, Role of Government and various bodies, Public Policy, Need and Role of Health Insurance, Latest development.

Unit III

Insurances for Hospital/ Medical establishments: Health Insurance: Uni-Heart care insurance, Trauma care group insurance, Coverages , Exclusions, limitations, terms and conditions. Liability Insurance: Meaning of Liability, special features of liability insurance, Professional Indemnity insurance for Doctors, Medical establishments and Liability insurance for hospitals, Coverages , Exclusions, limitations, terms and conditions.

Unit IV

Contemporary issues in Health Insurance: Foreign investment in Indian Insurance Sector, Insurance sector and supervisory approaches, Changing market structure and emerging opportunities, reforms and emerging economic environment for Health Insurance.

13 Suggested Resources for Reading

1. Stenens, W. S., *Health Insurance - Current Issues and Background*, Nova Science Publishers. 37
2. Mishra, M. N. and Mishra, S. B., *Insurance - Principles and Practice*, Sultan Chand and Sons.
3. Mahundran, T., *Health Insurance Sector in India*, Abhijit Publications.
4. Sorell, T., *Health Care, Ethics and Insurance*, Routledge Publications

Note: Latest edition of text books may be used.

1	Course Title	Healthcare Delivery System
2	**Course Code**	**HAM 703**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	**The main objectives of this course are to:** 1. Make the students understand the Health care Delivery System. 2. Inculcate the concept of Disaster Management, Biomedical waste Management and usage of computers in medicine.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the overview of health care delivery system. CO2. Analyze the clinical terminology in specialty wise. CO3. Evaluate the organization of Medical records department. CO4. Assess the importance of disaster management system, Medical insurance and Biomedical waste management in hospital. CO5. Analyze the computer applications in Medicine.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Introduction to Healthcare Delivery System: Healthcare Delivery System – An Overview – Common Terms of Healthcare Management: Terms Related to Levels of Health Care - Primary, Secondary and Tertiary - Systems of Medicine - Preventive and Curative Medical Care – General and Specialty Hospitals Indian Systems of Medicine.

UNIT– II

Specialty-wise Terminology: Specialty-wise Terminology – Pathology Terms of Common Use – Clinical, Diagnostic and Therapeutic Terms – Pediatric Services – Dental – Psychiatric – Casualty and Emergency – Neurology – Obstetrics and Gynecology – Dermatology Intensive Care – Coronary Care Services.

UNIT– III

Medical Records: Medical Records – Admission – Billing - Nursing Records - Diagnostic Records – Infection Control Records - Maintenance of Operation Theatres Records - Maintenance of Intensive Care Units Records - Clinical Records – Housekeeping Records – Food Records – Engineering Records– Maintenance Records – Security Records – Fatal Documents – Mortuary Maintenance Records – Transportation – Medico Legal Records

UNIT-IV

Disaster Management, Medical Insurance , Biomedical Waste Management: Disaster Management Basic Concepts - Principles of Disaster Planning - Objectives of a Hospital Disaster Plan - Disaster Committee - Organisation - Roles and Responsibilities Organising Disaster Facilities.Medical Insurance : Insurance Policy – Health Insurance in Indian Scenario – Economics of Life and Health Insurance - Importance – Health Insurance Policies – Administration of Health Insurance Schemes.Biomedical Waste Management & Handling Rule Management – Schedules – Segregation – Bio Medical Waste Collection – Transportation – Disposal - Modern Technology for Handling BMW -Radioactive Waste Handling.

13 Suggested Resources for Reading

1. Sharma D.K, Goyal R.C, 2013 . Hospital Administration and Human Resource Management. Prentice Hall of India (P) Ltd, New Delhi, 5th Edition.
2. Goel.S.L, 2001. Health Care System and Management. Deep &Deep Publications Pvt Ltd, New Delhi , 1st Edition.
3. R.D.Lee, 2005, Computers in Medicine, Tata McGraw-Hill, New Delhi, 1st Edition.

Note: Latest edition of text books may be used.

1	Course Title	International Classification Of Diseases
2	**Course Code**	**HAM 704**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The main objectives of this course are to: 1. Make the students understand the international classification of Diseases. 2. Understand the importance of using ICD in Mental and Health Statistics.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the history and uses- ICD design and structure. CO2. Analyze Basic coding guidelines and principles. CO3. Evaluate the Diseases Indexing and Operations. CO4. Evaluate Amendments of ICD 9 and ICD 10. CO5. Analyze the role of electronic medical records and its importance.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Basics of Classification: International Classification of Disease: History and Uses-Design of ICD and Structure.

UNIT– II

Principles of ICD: How to Use ICD - Basic Coding Guidelines - Principles and Rules of Coding - WHO Guidelines.

UNIT– III

Disease Index 9 hours International Classification of Procedures in Medicines-Indexing Of Diseases and Operations. ICD – 10 Salient Features of the Amendments to the ICD-10. Difference between ICD-9 And ICD-10.

UNIT-IV

Essentials of ICD: Role of International Classification of Diseases in Managing Electronic Medical Records. Importance of Using ICD in Mental and Health Statistics.

13 Suggested Resources for Reading

1. Mogli G.D, Medical Records-Organization & Management, New Delhi, Jaypee Brothers. 2 nd Edition.
2. G.D Kunders, 2008. Hospitals-Facilities, Planning & Management, Tata McGraw Hill, NewDelhi.
3. K. Park. 2019, Preventive and Social Medicine. Banarsidas Bhanot Publishers, 25th Edition.

Note: Latest edition of text books may be used.

1	Course Title	Total Quality Management Tools And Techniques
2	**Course Code**	**HAM 705**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Quality control Manager
8	**Course Objectives**	**The main objectives of this course are to:** 1. Make the students understand the concept of TQM Tools and Techniques. 2. Understand the various benchmarking and process re-engineering in Quality system.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the concept of Benchmarking system. CO2. Analyze various Benchmarking process and its implementation. CO3 .Evaluate the Quality Functional Deployment and its benefits. CO4. Determine Organizational Re-engineering framework and impact assessment. CO5. Analyze the statistical process control and its development.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Introduction to Benchmarking: Benchmarking – Introduction – Process of Benchmarking – Benefits – Pitfalls – Success Indicators – Gap Analysis Process – Benchmarking Template – Initiating the Process of Benchmarking. Quality Function Deployment: Quality Function Deployment – QFD Concept – QFD Process – QFD Team – Benefit of QFD – Voice of the Customer – Organisation of Information

UNIT– II

Benchmarking Process: Benchmarking Process – What to Benchmark – Benchmark Analysis – Determining Current Competitive Gap – Projecting Future Performance Levels - Integration – Developing Action Level – Implementing and Monitoring – Rehabilitation - Maturity - Beyond Benchmarking.

UNIT– III

Reengineering: Organisational Reengineering – Conceptual Framework Process – Process Re-Engineering- Bpr Philosophy – Possibilities and Pitfalls – Bpf Framework – Opportunity Assessment – Impact Assessment – Planning - Implementation of the Transition.

UNIT-IV

Statistical Process Control 7 hours Statistical Process Control : History – Development - Statistical Fundamentals - Pareto Diagram - Process Flow Diagram - Cause and Effect Diagram - Check Sheets - Histogram –Various Control Charts – Different Control Charts for Variable State of Control – Scatter Diagrams – Problems.

13 Suggested Resources for Reading

1. Dale H.Bester field, 2011. Total Quality Management. Pearson Education India ,New Delhi. 3 rd Edition.
2. S.K.Joshi, 2014. Quality Management in Hospitals. Jaypee Brother Medical Publishers (P) Ltd,New Delhi, 2nd Edition.
3. Sridhar Bhat, 2010. Total Quality Management. Himalaya Publishig House , Mumbai ,2nd Edition.

Note: Latest edition of text books may be used.

1	Course Title	Healthcare In India
2	**Course Code**	**HAM 706**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	**The main objectives of this course are to:** 1. Provide knowledge in Health sectors in India. 2. Learn and analyze the different system of Medicine. 3. Analyze the various medical administrative structure in India and recent trends in Hospital Administration.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Identify the Evolution and Development of Health care System. CO2. Compare the various system of medicines and its operations. CO3. Evaluate the administrative structure and various functions of Health Ministry Department. CO4. Explain the Medical Infrastructure Hospitals. CO5. Create and review of different reports on Health Care and recent trends in Hospitals
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT I

Health Sector in India: Development of Health Sector in India- Evolution of Health Care System - Medicine in Antiquity - Pre-Colonial Period – Colonial Period - Present Period-The Status of Indian Health. System of Medicine 9 Hours System of Medicine- Ayurveda – Yoga – Naturopathy – Siddha – Unani – Homeopathy – Modern Allopathic medical system.

UNIT II

Administrative structure: Administrative Structure in National Level - Health Minister of State - Deputy Ministers – Departments under Health Ministry - Role, Duty and Functions.

UNIT III

Medical Care Infrastructure: Prevention of Food and Adulteration Act 1954 – Bio-Medical Waste (Management and Handling) Rules 2011 – Pollution Control Acts and Regulations of India 1981 – Indian Boilers Act 1923.

UNIT IV

Reports on Health Care: Review of Different Reports on Health Care in India - Recent Trends in Hospital Administration.

13 Suggested Resources for Reading

1. Kunders G.D. 2007, Hospitals Facilities Planning and Management. Tata McGraw-Hill Publishing company ltd, New Delhi. 1st Edition.
2. Goyal R.C, Hospital Administration and Human Resource Management. Prentice Hall of India (P) Ltd, New Delhi. 5th Edition.
3. Goel.S.L.,Health Care Organization and Structure. Deep &Deep Publications Pvt Ltd, New Delhi.

Note: Latest edition of text books may be used.

SEMESTER VIII

1	Course Title	Service Marketing
2	**Course Code**	**HAM 801**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	The objective of this course is to enhance the marketing skills of the student with special reference to Hospital Services marketing.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand and explain the nature and scope of services marketing and present about this in a professional and engaging manner. CO2. Provide a theoretical and practical basis for assessing service performance using company examples and report on this in a professional, logical and coherent way. CO3. Identify and discuss characteristics and challenges of managing service firms in the modern world. CO4. Discuss key linkages between marketing and other business functions in the context of designing and operating an effective service system. CO5. Work effectively and efficiently in a team addressing services marketing topics.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Fundamentals for Service Marketing: Introduction, Types of Services, Characteristics of Services, Consumer Behavior in Services, Service Marketing

Unit II

Consumer Expectations of Services, Customer Perception of Services, Building Customer Relationship. Service Recovery, Service Innovation and Design.

Unit III

Employees and Customers role in service delivery. Role of Intermediaries and Electronic Channels. Managing Demand and Capacity.

Unit IV

Managing Service Promises, Pricing Strategies. The Financial and Economic Impact of Services**.**

13 Suggested Resources for Reading

1. Kotler P., *Marketing Management (Millennium Edition)*, PHI.
2. Zeithaml, V. and Bitner, M. J., *Services Marketing: Cases in Marketing Management*, McGraw Hill.
3. Srinivasan, R., *Services Marketing (The Indian context)*, Prentice-Hall India.
4. Bhattacharya, C., *Services Marketing*, Excel Books.
5. Shankar, R., *Marketing Management*, Tata Mc Graw Hill.
6. Still, R. R., Cundiff, E. W., and Govani, N. A. P., *Sales Management,* PHI.
7. Bookman, M. Z. and Bookman, K. R., *Medical Tourism in Developing Countries*, Palgrave Macmillan

Note: Latest edition of text books may be used.

1	Course Title	Business Environment
2	**Course Code**	**HAM 802**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Relation Executive
8	**Course Objectives**	**The Main objective of course are:** 1. To acquaint students with the issues of domestic and global environment in which business has to operate. 2. To relate the Impact of Environment on Business in an integrated manner. 3. To give an exposure to important commercial and industrial laws.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the different environments in business climate. CO2. know the minor and major factors affecting the business in various streams. CO3. Know the different environments like political, technological, and economic environment in the business. CO4. Acquire in-depth knowledge about legal environment.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT – I

Dynamics of Business and its Environment – Technological, Political, Social and Cultural Environment - Corporate Governance and Social Responsibility - Ethics in Business – Economic Systems and Management Structure - Family Management to Professionalism - Resource Base of the Economy - Land, Forest, Water, Fisheries, Minerals - Environmental Issues.

UNIT - II

Infrastructure - Economic- Social, Demographic Issues, Political context - Productivity Factors, Human Elements and Issues for Improvement - Global Trends in Business and Management - MNCs - Foreign Capital and Collaboration - Trends in Indian Industry - The Capital Market Scenario.

UNIT - III

Law of Contract - Agreement - Offer - Acceptance - Consideration - Capacity of Contract Contingent Contract - Quasi Contract - Performance - Discharge - Remedies to breach of Contract - Partnership - Sale of Goods - Law of Insurance - Negotiable Instruments - Notes, Bills, Cheques - Crossing - Endorsement - Holder in due course - Holder in value - Contract of Agency.

UNIT - IV

Company - Formation - Memorandum - Articles - Prospectus - Shares - Debentures -Directors - Appointment - Powers and Duties - Meetings - Proceedings – Management - Accounts - Audit - Oppression and Mismanagement - Winding up. Factory Act – Licensing and Registration of Factories, Health, Safety and Welfare measures Industrial Disputes Act – Objects and scope of the Act, Effects of Industrial Dispute, Administration under the Act- Minimum Wages Act - Workmen Compensation Act.

13 Suggested Resources for Reading

1.Pathak, LEGAL ASPECTS OF BUSINESS, Tata McGraw- Hill Publishing Company Limited, New Delhi, 2010.

2. Keith-Davis & William Frederick, BUSINESS AND SOCIETY, McGraw-Hill, Tokyo.

3. M.M. Sulphey & Az-har Basheer, LAWS FOR BUSINESS, PHI Learning Pvt. Ltd. New Delhi, 2011

4. Maheswari & Maheswari, MERCANTILE LAW. Himalaya Publishing House. Mumbai

5. Rudder Dutt & Sundaram, INDIAN ECONOMY, Vikas Publishing House, New Delhi.

6. Veena Keshav Pailwar, ECONOMIC ENVIRONMENT OF BUSINESS, PHI Learning Pvt. Ltd, New Delhi, 2010

Note: Latest edition of text books may be used.

1	Course Title	Operation Research & Management
2	**Course Code**	**HAM 803**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	1. To familiarize the Operations Management concepts. 2. To introduce various optimization techniques with managerial perspective. 3.To facilitate the use of Operations Research techniques in managerial Decisions.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand fundamental concepts and principles of management on emerging trends and developments in management. CO2. Apply organizational planning systems and Decision making in health care. CO3. Examine various types of organizations and its elements. CO4. Propose controlling techniques for health care organizations.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT –I

Introduction to Operations Management - Process Planning - Plant Location - Plant Lay out -Introduction to Production Planning.

UNIT –II

Stages of Development of Operations Research- Applications of Operations Research- Limitations of Operations Research- Introduction to Linear Programming- Graphical Method- Simplex Method - Duality.

UNIT-III

Transportation Problem- Assignment Problem - Inventory Control - Introduction to Inventory Management - Basic Deterministic Models - Purchase Models - Manufacturing Models with and without Shortages.

UNIT-IV

Shortest Path Problem - Minimum Spanning Tree Problem - CPM/PERT - Crashing of a Project Network. Game Theory- Two Person Zero-sum Games -Graphical Solution of (2 x n) and (m x 2) Games - LP Approach to Game Theory - Goal programming - Formulations - Introduction to Queuing Theory - Basic Waiting Line Models: (M/M/1):(GD/a/a), (M/M/C):GD/a/a).

13 Suggested Resources for Reading

1. Kanishka Bedi, PRODUCTION & OPERATIONS MANAGEMENT, Oxford, NewDelhi, 2007
2. Panneerselvam, R, OPERATIONS RESEARCH, Prentice-Hall of India, New Delhi, 2002.
3. G.Srinivasan, OPERATIONS RESEARCH, PHI Learning, NewDelhi,2010
4. Tulsian & Pandey, QUANTITATIVE TECHNIQUES, Pearson, NewDelhi, 2002
5. Vohra, QUANTATIVE TECHNIQUES IN MANAGEMENT, Tata McGrawHill, NewDelhi, 2010

Note: Latest edition of text books may be used.

1	Course Title	Hospital Facilities Management
2	**Course Code**	**HAM 804**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40TH+40PR+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objectives of this course is to enable the students to get insights into various aspects like importance, functions, policies and procedures of various non clinical facilities.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** get familiarized with support service systems **CO2.** get acquainted with management of services and facilities in hospital.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT I

Nutrition and Dietary services – Pharmacy services – Medical Records services.

UNIT II

Facilities Engineering – Maintenance of Civil Assets – Electrical supply and Water supply – Medical gas pipeline – Plumbing and Sanitation – Air conditioning system – Hot water and Steam supply – Communication Systems – Biomedical engineering departments in modern hospitals.

UNIT III

Laundry services – House keeping services – CSSD-Energy conservation methods – AMC. Ambulance services – Mortuary services – Hospital security services.

UNIT-IV

Disaster management – Fire hazards – Engineering Hazards – Radiological hazards.-Outsourcing of Support services –Waste disposal and management - few case studies.

13 Suggested Resources for Reading

1. G.D.Kunders, HOSPITAL AND FACILITIES PLANNING AND DESIGN
2. Jacob Kline, HAND BOOK OF BIO-MEDICAL ENGINEERING
3. Webster J.G and Albert M. Coo, CLINICAL ENGINEERING PRINCIPLES AND PRACTICES
4. Antony Kelly, MAINTENANCE PLANNING AND CONTROL

Note: Latest edition of text books may be used.

1	Course Title	Bio Medical Instrumental
2	**Course Code**	**HAM 805**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH+20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Quality Control Manager
8	**Course Objectives**	The objective of this course is to cover the Operations and Maintenance aspects with reference to minimum Utilization of resources in a hospital.
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1.Understand the applications of electronics in diagnostic and therapeutic area. CO2.control the quality of instruments used in patient care. CO3.effectively manage the medical instruments contributing to patient safety.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT I

List of common medical equipments, Justification of purchase proposal, hospital need assessment, Equipment selection guideline, estimation of cost and planning, purchase, installation, commissioning, Replacement and buy back policy, International and indigenous standards.

UNIT II

Operations Strategy: Operations strategy a competitive tool, elements, technology selection and process: development, developing operations strategy. operations of clinical services, supportive services, and administrative services general introduction to the various specialties, super specialties and other subspecialties.

UNIT– III

Value Management: Value engineering, value analysis, quality control, applications in hospital. Technology Management in Hospitals. Evolution of technology in hospitals, advanced technology in diagnostics and therapeutics, telemedicine concepts and applications, artificial intelligence and robotics in Healthcare.

UNIT-IV

Maintenance Management: Objectives, types of maintenance systems, equipment maintenance, quality and reliability, equipment history and documents, maintenance planning, maintenance information system, maintenance and monitoring of biomedical equipments, predictive maintenance, equipment availability, spares management, replacement policy, depreciation and loss of value, economic life, costing, cost of stand by, maintenance in hospital. Bio-Medical Technology, application in hospital environment, calibration tests, maintenance features, hazards.

13 Suggested Resources for Reading

1. Medical Technology, application in hospital environment, calibration tests, maintenance of hazards Srinivasan A.V. (ed), Managing a modern hospital, Chapters 12, Response Books, New Delhi, Barry, Jay Hazier, Principles of Operations Management, Prentice Hall, New Jersey
2. Roger G., Operations Management - Decision Making in Operations Function, rawHiII., New Delhi.
3. , Elwood S. and Sarin, Rakesh K., Modern Production/Operations Management, John Wiley illS, Singapore, 1987.
4. Lee J. and Larry p. Ritzman, Operations Management - Strategy and Analysis, 'ton - Wesley Publications.
5. International Journal of Operations and Quantitative Management, (IJOQM) released by Nirma Institute of Management – Ahmedabad

Note: Latest edition of text books may be used.

1	Course Title	Newer Developments in Hospital Administration
2	**Course Code**	**HAM 806**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of the course is to familiarize the students with recent advances in the administration of healthcare services.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** 1. Appreciate the concept and its applications of eHealth, m-health in the efficient delivery of hospital services. 2. Design strategy to promote Medical Tourism in hospital 3. design disaster preparedness and response plan for the hospital. 4. promote electronic health record in the hospital
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

eHealth: Telemedicine, mHealth, electronic appointment & follow up system ,Telemedicine: scope, objective & functions, Classification, Global Scenario, Indian Scenario, Challenges. Growth of mobile phone use and potential of mobile health, mobile health & its application, calling, Short message service, Newer wireless devices & technology - Patient keeper, Hippocrates Rx for android, Motion Computing C5 Mobile Clinical| Assistant, Treatment compliance monitoring**.**

UNIT– II

Electronic medical record system, EMR access & updation, electronic appointment & follow up.

UNIT– III

Hospitality Services in Hospital: Patient as Guest, Aims and objectives of Hospitality Management, Role of Hospitality Management, Methods of Hospitality Management in a hospital, Attractive look, Effective conversation, Multi lingual, Smart dress, Etiquette and manners. Medical Tourism: Introduction, Health & Medical tourism, Efforts and features of the hospitals to facilitate medical tourism, Client perspective of service delivery, Implications of medical tourism.

UNIT-IV

Hospitals in a globalized World: Direct & indirect effect of globalization on healthcare systems and services, globalization of medical services, Patients in a globalized world, Role of teaching & research hospitals in improving global health.

13 Suggested Resources for Reading

1. Hospital Administration by S. L. Goel
2. Quality Management by Janakiraman & Gopal
3. Medical Records by Geeta Sahay
4. Ethical issue related to Medical Profession by NIHFW, Monograph, Govt. of India
5. Medical Tourism: Adventures in India by Steven F. Larson (2013)

Note: Latest edition of text books may be used.

1	Course Title	Drug Management
2	**Course Code**	**HAM 807**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours of lectures for Theory + … hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH +20IA) 1. The exam of this course will be of 3hours duration. 2. There will be no objective type questions. 3. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. 4. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Care Coordinator
8	**Course Objectives**	The objective of this course is to familiarize the students with different aspects of drug management and quality control in healthcare
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. have profound knowledge of materials and their medical applications. CO2. organize and manage pharmaceuticals. CO3. effectively manage processes relating to purchasing, stocking, and distribution of pharmaceuticals in hospitals. CO4. assure quality in the hospital drug inventory and pharmacy.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Drug Management ; Present scenario, Hospital Pharmacy Licenses, Drug Licenses, Narcotics drugs

UNIT– II

Purchase of drugs and other consumable materials. Drug Storage, Inventory Control – ABC, VED, SDE, FSN Analysis

UNIT– III

Rational use of drugs and Prescription Audit, Spurious drugs, Banned drugs

UNIT-IV

Methods of ordering –a) Two bin system (Lead Time, Buffer stock, Reorder Level), b) Cyclic System, Pharmacy Billing. Computerized drug management system, Procedure of Drug Indenting, On time Drug dispensing.

13 Suggested Resources for Reading

1.Materials Management – Gopalakrishnan
2. Matgerials Management - A. C. Dutta

Note: Latest edition of text books may be used.